

Assignment 3: A Complete Spoken-Video Translation Pipeline

Preserving speaker voice characteristics and synchronizing lip movements

Mohamed Kitaz & Abdulaziz Kitaz
Johannes Kepler University Linz

Component	Implementation / Result
Input sample	id0001, English video, h_vamIjcIHE/00000.avi
Target language	German
Speech recognition	Whisper detected English and produced transcription
Voice generation	XTTS voice-cloned German audio, duration matched to 91.36 seconds
Lip synchronization	Wav2Lip produced final translated video
Embedding extraction	ECAPA-TDNN audio (192-dim), FaceNet visual (512-dim)
Baseline result	Original and generated samples both classified correctly as label 0
Additional metric	Generated German sample confidence margin = 0.9119

Abstract

This report presents the implementation of a complete pipeline that converts a spoken input video into a target language while preserving speaker identity and synchronizing the mouth movement with the generated speech. The pipeline was demonstrated on one English video from identity id0001 and generated a German lip-synchronized output video. Audio and visual embeddings were extracted from both the original and generated samples using ECAPA-TDNN and FaceNet. The trained FOP baseline classifier was then used to compare recognition performance. In addition to cosine similarity, a classifier confidence margin was added as a task-specific identity-preservation metric. Finally, a synthetic-data analysis was performed to estimate how many generated samples would be required to balance the training identities.

1. Introduction

The goal of Assignment 3 is to build a full pipeline for spoken-video translation. The system should take a video with speech as input, detect and transcribe the spoken language, translate the text into a target language, generate the translated speech while keeping the speaker voice characteristics, and finally synchronize the lips with the new audio. After generation, embeddings are extracted from original and generated audio-visual data and evaluated using the baseline model.

The implementation follows the same practical style used in the previous baseline and SBNet reports: clear task-based sections, implementation details, result tables, and short analysis of the main numbers.

2. Task 1 - Pipeline Creation

The implemented pipeline consists of seven main stages. The original video is first converted into a WAV audio file using FFmpeg. Whisper is then used for language detection and speech recognition. The recognized English text is translated into German. A short clean speaker prompt is cut from the original audio, and XTTS uses this prompt to synthesize German speech with a similar speaker voice. The generated German audio is then duration-matched to the original video length. Finally, Wav2Lip generates a lip-synchronized German video.

Stage	Tool / Method	Output
Audio extraction	FFmpeg	sample_original.wav
Lang. detect + ASR	Whisper	Detected language: English; transcribed text
Translation	deep-translator	German translated text
Speaker prompt	FFmpeg	20-second crop sample_speaker_prompt_20s.wav
Voice cloning and TTS	XTTS v2	sample_generated_de_full_safe.wav
Duration matching	Padding/trimming in Python	sample_generated_de_full_safe_matched.wav
Lip synchronization	Wav2Lip-GAN	sample_lipsync_full_de.mp4

File	Duration / Details
Original audio	91.36 seconds
Generated German audio	88.52 seconds
Matched German audio	91.36 seconds
Final lip-sync video	about 91.24 seconds, 25 FPS, 112 x 112 output

The generated German audio was slightly shorter than the original audio, so silence padding was added to match the original duration. This step is important because Wav2Lip expects the face video and speech audio to have compatible timing.

3. Task 2 - Embedding Extraction

After generating the translated video, embeddings were extracted from both original and generated data. For the audio modality, ECAPA-TDNN was used through the SpeechBrain speaker-recognition model. For the visual modality, FaceNet/InceptionResnetV1 was used. Since the original and generated videos are already face-centered, 32 evenly spaced frames were sampled from each video and averaged to form one video-level visual embedding.

Modality	Encoder	Original shape	Generated shape	Cosine similarity
Audio	ECAPA-TDNN	(192,)	(192,)	0.6962
Visual	FaceNet / InceptionResnetV1	(512,)	(512,)	0.9805

The visual cosine similarity is very high (0.9805), which means the generated lip-synced video keeps almost the same face identity representation as the original video. The audio similarity is lower (0.6962), which is expected because the generated speech is synthetic and translated into German. However, the audio embedding still preserves a meaningful amount of speaker information.

Why keep cosine similarity?

Cosine similarity is useful at the embedding level because it directly measures how close the original and generated representations are in the feature space. However, it does not fully answer whether the classifier still recognizes the same identity. Therefore, Task 3 adds a classifier-based confidence margin as a more task-specific identity-preservation metric.

4. Task 3 - Baseline Evaluation and Comparison

The FOP baseline model from the previous assignment was used for comparison. The baseline uses separate face and voice branches and performs closed-set speaker classification over 36 identities. The generated A3 embeddings were converted into the same format used by the baseline: 192-dimensional ECAPA audio features and 512-dimensional FaceNet visual features.

The sample identity id0001 corresponds to label 0 in the baseline data. Two rows were created for evaluation: one for the original English sample and one for the generated German sample. The corrected model output was selected from output[1], which contains the classifier logits with shape [batch, 36].

Sample	True label	Predicted label	Confidence	Correct
Original English	0	0	0.9999	Yes
Generated German	0	0	0.9541	Yes

The baseline correctly classified both the original and generated samples as identity id0001. The confidence decreased from 0.9999 to 0.9541 after synthetic generation. This decrease is reasonable because the generated sample contains synthetic German speech and Wav2Lip-generated mouth motion. Still, the prediction remained correct, showing that the pipeline preserved enough audio-visual identity information for the classifier.

Additional identity-preservation metric: confidence margin

In addition to cosine similarity, the classifier confidence margin was computed. The confidence margin is defined as the probability of the correct identity minus the probability of the second-best identity. This metric is more task-specific than cosine similarity because it measures whether the generated sample is still clearly recognized by the identity classifier.

Sample	Top-1 probability	Second-best probability	Confidence margin
Original English	0.9999	0.0000	0.9999
Generated German	0.9541	0.0422	0.9119

For the generated German sample, the confidence margin was 0.9119. This shows that the correct identity remained strongly separated from the nearest competing identity. Therefore, even though the synthetic generation slightly changed the audio-visual representation, the sample was still clearly recognized as id0001 by the baseline classifier.

5. Task 4 - Synthetic Data Analysis

For the bonus task, the number of real training samples was counted for each identity in the English and German training splits. The dataset is strongly imbalanced. The smallest identities contain only 3 real samples, while the largest identity contains 155 samples. The mean number of real samples per identity is 37.83 and the median is 31.

A practical balancing target of 50 samples per identity was selected. This value is close to the upper part of the distribution and avoids forcing every identity to match the largest identity. The number of required synthetic samples was computed with the following rule:

```
synthetic_needed = max(0, 50 - current_real_samples)
```

Statistic	Value
Number of identities	36
Minimum real samples	3
Maximum real samples	155
Mean real samples	37.83
Median real samples	31
Recommended target per identity	50
Total synthetic samples required	708

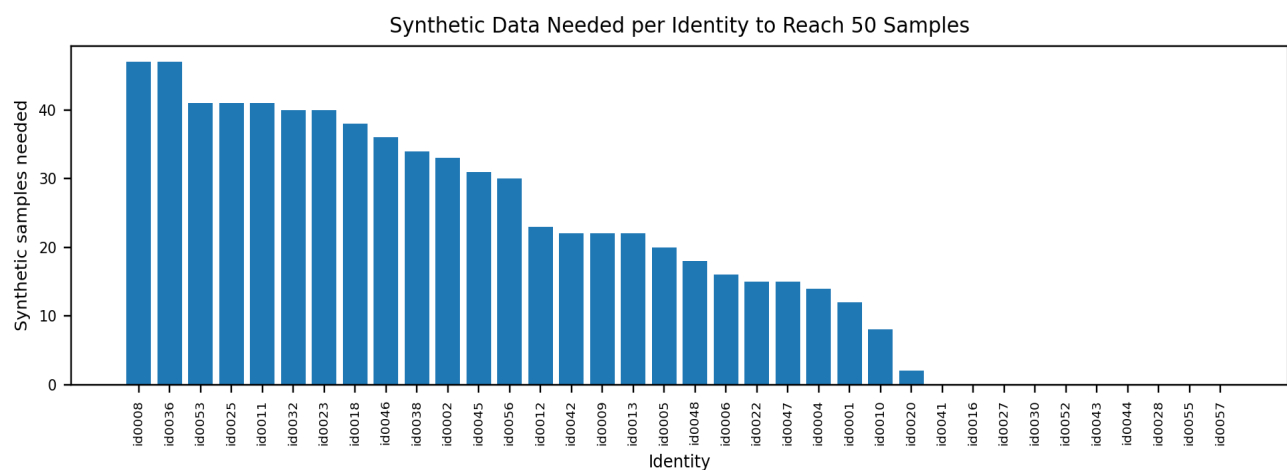


Figure 1. Synthetic samples needed per identity to reach 50 training samples.

Identity	Real samples	Synthetic needed
id0008	3	47
id0036	3	47
id0053	9	41
id0025	9	41
id0011	9	41
id0032	10	40
id0023	10	40
id0018	12	38
id0046	14	36
id0038	16	34

The analysis shows that synthetic generation should focus on the identities with the fewest real samples. Generating the same amount of synthetic data for all identities would be inefficient and could increase imbalance for already frequent identities.

6. Conclusion

The complete A3 pipeline was successfully implemented. The system transcribed the original English video, translated the text into German, generated German speech using the original speaker voice as a prompt, matched the generated audio length to the original video, and synchronized the mouth movements with Wav2Lip. ECAPA-TDNN and FaceNet embeddings were extracted from both original and generated data. The generated sample remained recognizable by the FOP baseline, with correct prediction of identity id0001 and confidence 0.9541. The additional confidence-margin metric showed a strong generated-sample margin of 0.9119, confirming that the correct identity remained clearly separated from competing identities. Finally, the synthetic-data analysis showed that 708 generated samples would be needed to balance the training set to 50 samples per identity.

7. Important Output Files

Output	Path
Final translated lip-synced video	outputs/video_lipsync/sample_lipsync_full_de.mp4
Task 2 embedding summary	outputs/task2_embedding_summary/task2_embedding_summary.csv
Corrected baseline results	outputs/task3_baseline_results/a3_baseline_original_vs_generated_results_CORRECTED.csv
Confidence margin results	outputs/task3_baseline_results/a3_baseline_results_with_confidence_margin.csv
Synthetic data analysis	outputs/task4_synthetic_data_analysis/synthetic_needed_per_identity.csv
Synthetic data plot	outputs/plots/task4_synthetic_needed_per_identity.png

References

- [1] Assignment 3 instructions: complete spoken-video translation pipeline with Whisper, translation, voice cloning/TTS, lip synchronization, ECAPA-TDNN, and FaceNet.
- [2] Previous FOP baseline report and SBNet report used as formatting/style references and as the baseline background.